VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI



A project report
on

"IOT and Machine learning Based Smart Health Monitoring Belt with ECG using ESP32 "

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING

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Under the guidance of

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

PES INSTITUTE OF TECHNOLOGY AND MANAGEMENT

NH-206, Sagar Road, Shivamogga – 577 204

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Institute Vision

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Institute Mission

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Department Vision

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Department Mission

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- To develop a framework for collaboration and multidisciplinary activities to ensure ethical and value based education to address social needs.

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Certificate from the Institute

This is to certify that the project work was carried out jointly by

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are bonafide students of the Bachelor of Electronics and Communication Engineering program of the Institute (2023–2027 Batch), affiliated to Visvesvaraya Technological University, Belagavi.

A project report entitled **"IOT and Machine learning Based Smart Health Monitoring Belt with ECG using ESP32"** has been submitted by them under the guidance of Mrs.Sangeetha B S, Assistant Professor, Department of Electronics and Communication Engineering , PESITM, Shivamogga, in partial fulfillment of the requirements for the award of the degree of Bachelor of Electronics and Communication Engineering.

Guide Name	**HoD Name**	**Principal Name**
Designation	**Designation**	**Designation**

Viva-voce Examination Date: _____

Signature of Internal Examiner

Signature of External Examiner

DECLARATION

We, **Aishwarya T S, Ankitha K, Bhoomika R** and **Mayur C** hereby declare that the project report entitled **"SOT and Machine learning Based Smart Health Monitoring Belt with ECG using ESP32"** has been carried out by me under the guidance of Mrs.Sangeetha B S , Assistant Professor, Department of ECE, PESITM, Shivamogga.

I further declare that this project report is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi.

I also declare that this report is based on the original work undertaken by me and has not been submitted for the award of any degree or diploma from any other University/Institution.

Place: Shivamogga
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ACKNOWLEDGEMENT

We take this opportunity to express our deep sense of gratitude to our guide **Mrs. Sangeetha B S**, Assisstant professor, Department of ECE, PESITM, Shivamogga for his kind support, guidance, and encouragement throughout the course of this dissertation work.

We are highly obliged to **Dr. Vishnu V M**. Professor and Head, Department of ECE, PESITM, Shivamogga for his kind support and encouragement throughout the course of this work.

We are highly grateful to **Dr. Swamy D R Principal**,**Dr. Sekar R Vice Principal** PESITM, Shivamogga for providing us an opportunity to fulfill our most cherished desire of reaching the goal.

Finally, we are thankful all the teaching and non-teaching staff, my parents and friends, who helped us in one way or the other throughout our project work.

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Abstract

The proposed project is an IOT and Machine learning based Smart Health Monitoring Belt designed to provide continuous, portable, and intelligent monitoring of important physiological parameters. The system integrates an ECG sensor, SpO sensor, and body-temperature sensor with an ESP32 microcontroller to collect real-time health data from the user. The measured parameters are processed by the ESP32 and displayed locally on an OLED display, allowing the user to observe vital signs without requiring a smartphone or computer. At the same time, the collected data is transmitted through Wi-Fi to the ThingSpeak cloud platform, where it can be stored, visualized, and monitored remotely. A dedicated web application is incorporated to maintain the user's profile, including additional parameters such as age, gender, height, weight, existing health information, and other relevant personal parameters. These profile parameters are combined with the sensor measurements to create a more comprehensive input for a machine-learning model. The ML system analyzes patterns in ECG, SpO, temperature, and user-profile information to classify or predict the user's overall health status, such as Normal, Attention Required, or Critical/Risk. The system is intended as a low-cost prototype for continuous health monitoring and early indication of abnormal conditions rather than as a replacement for certified medical equipment. The combination of wearable sensing, cloud connectivity, web-based profile management, and machine learning provides an intelligent framework for remote and personalized health-status.

The project outcomes demonstrate improved task completion rates and user motivation compared to existing tools. In addition, the work contributes to Sustainable Development Goal (SDG) 4: Quality Education by promoting effective learning practices, and SDG 9: Industry, Innovation, and Infrastructure through the use of modern digital technologies.

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Chapter 1

Introduction

Continuous monitoring of physiological parameters is becoming increasingly important because many health abnormalities can develop gradually and may not be recognized immediately by the individual. Conventional health monitoring generally requires dedicated medical instruments and periodic visits to healthcare facilities, making continuous observation inconvenient for many users. A wearable smart belt can provide a more convenient approach by integrating multiple physiological sensors into a compact wearable platform. In this project, an ESP32 acts as the central controller because it provides sufficient processing capability, multiple interfaces for sensors, and built-in Wi-Fi connectivity for cloud communication. An ECG sensor is used to acquire electrical activity of the heart, while an SpO sensor measures blood oxygen saturation and pulse-related information. A temperature sensor measures body temperature, providing another important indicator of the user's physiological condition. These measurements are continuously processed and presented on an OLED display attached to the belt, enabling immediate access to the measured values. The project further extends conventional wearable monitoring by incorporating cloud computing and machine learning. Instead of limit the system to displaying individual sensor readings, the ESP32 can transmit measurements to ThingSpeak through Wi-Fi, enabling historical data storage and visualization. The web application provides a user profile containing additional parameters that can influence health-status interpretation. For example, the same physiological reading may have different significance depending on factors such as age, body characteristics, and other profile information. Therefore, the proposed ML model can combine the real-time sensor parameters with selected profile parameters to produce a more personalized health-status prediction.

The system can continuously evaluate incoming measurements and provide a simplified status indication through the OLED and web interface. This architecture demonstrates how IoT, wearable electronics, cloud platforms, web applications, and machine learning can be integrated into a single health-monitoring solution.

1.1 Problem Statement

Traditional health monitoring systems often require separate devices and manual recording of physiological data, making continuous monitoring difficult. The proposed ESP32-based Smart Health Monitoring Belt integrates ECG, SpO_2, and temperature sensors into a single wearable system. The collected data is displayed in real time on an OLED display and uploaded to ThingSpeak for remote monitoring and storage. A web application maintains user profile information for personalized analysis. Machine learning is used to analyze physiological parameters along with user-specific information instead of relying only on fixed thresholds. The system provides an easy-to-understand health-status classification and supports early identification of abnormal measurements. It is designed for health monitoring and awareness, and not for independent medical diagnosis.

1.2 Project Description

The ESP32-Based Smart Health Monitoring Belt is a wearable system designed for continuous and portable monitoring of important physiological parameters. It uses ECG, SpO, and temperature sensors connected to an ESP32 to collect real-time health data. The measured values are displayed on an OLED display for immediate monitoring and transmitted through Wi-Fi to the ThingSpeak cloud for storage and visualization. A web application maintains user profile information such as age, gender, height, and weight. These profile details are combined with sensor data and analyzed using machine learning to estimate the user's health status. The system can classify the condition into categories such as Normal, Attention Required, or Risk/Critical. Overall, the project integrates embedded systems, wearable technology, IoT, cloud computing, web applications, and machine learning for intelligent health monitoring

1.3 Objectives

- To develop a wearable smart health-monitoring belt using ESP32, ECG, SpO_2, and temperature sensors for continuous acquisition of physiological parameters.
- To display real-time health parameters such as ECG, SpO_2, temperature, heart rate, and predicted health status on an OLED display.
- To transmit and store health data on the ThingSpeak cloud through the ESP32's built-in Wi-Fi connectivity for remote monitoring and historical analysis.
- To develop a web application for user-profile management and combine profile parameters such as age, gender, height, weight, and other selected attributes with sensor data.
- To implement a machine-learning model that analyzes sensor and profile parameters to predict the user's overall health status and identify potentially abnormal conditions.

1.4 Scope

The scope of this project includes the development of a wearable health-monitoring belt for real-time measurement of ECG, SpO_2, and body temperature. The ESP32 acts as the main processing and communication controller. An OLED display provides immediate visualization of the measured parameters, while Wi-Fi enables data upload to ThingSpeak for remote monitoring and historical analysis. A web application can be used to maintain user profiles and view current and previous health data. The ML component analyzes sensor readings along with user-specific information to estimate overall health status. The system can be expanded by integrating additional sensors such as blood pressure, motion, respiratory-rate, and fall-detection sensors. Future improvements may include larger datasets, personalized ML models, alerts, trend analysis, and caregiver notifications.

1.5 Purpose

The purpose of this project is to develop a wearable smart health-monitoring belt that continuously monitors important health parameters such as ECG,SpO, heart rate, and body temperature. The system uses an ESP32 to process the sensor data, display the readings on an OLED screen, and upload them to the ThingSpeak cloud for remote monitoring. It also uses machine learning to analyze sensor readings along with user-profile information and estimate the overall health status. The project aims to provide convenient and personalized health monitoring and support the early identification of potentially abnormal health conditions.

1.6 Relevance to Sustainable Development Goals (SDGs)

This project aligns with the following UN Sustainable Development Goals (SDGs):

- **SDG 3 – Good Health and Well-being:** The project supports continuous monitoring of ECG, SpO, heart rate, and temperature to promote better health awareness and early identification of abnormal conditions.
- **SDG 9 – Industry,Innovation and Infrastructur:** It promotes innovation by integrating wearable sensors, ESP32, IoT, cloud computing, web applications, and machine learning into a smart healthcare system .
- **SDG 10 – Reduced Inequalities:** low-cost and portable design can help make basic health-monitoring technology more accessible to people who may have limited access to conventional healthcare facilities.
- **SDG 12 – Responsible Consumption and Production:** The wearable system uses a compact, reusable electronic platform for continuous monitoring, supporting efficient use of resources and reducing dependence on multiple separate monitoring devices.

Chapter 2

Literature Review / Existing System

According to reference [1] presented a wearable ECG monitoring system combined with machine-learning techniques for continuous cardiac monitoring. The research addresses the increasing demand for portable cardiovascular monitoring caused by the growing prevalence of cardiovascular diseases. The proposed approach focuses on acquiring ECG signals using wearable hardware and applying computational methods to analyze cardiac activity outside conventional clinical environments. Machine-learning techniques are incorporated to improve the interpretation of ECG signals and assist in identifying cardiac abnormalities. The study demonstrates how wearable ECG acquisition can be combined with intelligent signal processing techniques to provide more accessible cardiac monitoring. Such systems can reduce the dependency on large clinical equipment and enable physiological information to be collected during normal daily activities. The work is particularly relevant to the proposed smart belt because it establishes the feasibility of combining wearable ECG sensing with machine learning for intelligent health assessment.

According to reference [2] This research proposes an ensemble deeplearning approach for remote patient monitoring using Internet of Things technology. The study combines Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks to analyze health-related information collected through IoT-based monitoring systems. The CNN component is useful for extracting meaningful patterns from physiological data, while LSTM networks are suitable for identifying temporal relationships in continuously acquired health signals. The research demonstrates the importance of combining multiple deep-learning architectures for comprehensive health assessment rather than depending on a single physiological measurement. The IoT architecture enables remote collection

and analysis of patient information, supporting continuous monitoring outside hospitals. The work provides an important foundation for the proposed smart belt because the proposed system can similarly combine multiple sensor measurements and use ML to classify overall health status.

According to reference [3] This introduces an intelligent healthcare monitoring framework called HealthEdge, combining IoT, edge computing, multiple physiological sensors, and deep-learning-based anomaly detection. The system considers a diverse collection of physiological parameters including respiratory rate, pulse rate, SpO, galvanic skin response, body temperature, and other health-related signals. Instead of sending all raw information directly to a remote cloud server, edge computing is used to support real-time analysis closer to the data source. This approach can reduce communication latency and allow potentially abnormal health conditions to be detected more quickly. The multi-sensor architecture is particularly relevant to wearable healthcare because health conditions are often better characterized using multiple physiological parameters rather than a single measurement. The study provides a strong basis for the proposed project, where ESP32 can collect ECG, SpO, and temperature data and cloud/edge ML techniques can be used for health-status prediction.

According to reference [4] Seddiki et al. developed an IoT healthcare architecture focused on real-time ECG analysis and comprehensive patient monitoring. The system uses deep-learning models trained using the MIT-BIH ECG dataset, with one model dedicated to R-R peak detection and another model designed to classify ECG signals into multiple disease categories. In addition to ECG analysis, the proposed IoT architecture collects physiological parameters such as ECG, SpO, and temperature using an ESP32 and Raspberry Pi-based platform. The collected information can be transmitted for remote analysis, thereby combining biomedical sensing with IoT communication and artificial intelligence. The study reported high classification performance for its ECG models and demonstrated the feasibility of integrating physiological sensors with intelligent analysis. This paper is particularly important for the proposed smart belt because it closely matches the combination of ECG, SpO, temperature, ESP32-based acquisition, and ML-based health analysis.

According to reference [5] This research presents an intelligent garment designed for real-time monitoring of individuals experiencing long COVID19 symptoms. The sys-

tem combines several physiological parameters, particularly heart rate, body temperature, SpO, and ECG signals, and uses explainable data-fusion techniques with multiple machine-learning models to assess health status. Rather than depending on a single parameter, the approach combines physiological information to identify patterns associated with the health condition. The study reported an accuracy of approximately 97.5 percent and identified SpO and ECG ST-interval characteristics as particularly influential indicators in its experimental setting. The intelligent garment concept is highly relevant to a smart belt because both systems demonstrate how physiological sensors can be embedded into wearable clothing to facilitate continuous monitoring. The work also supports the proposed idea of using multiple physiological parameters together with ML instead of relying solely on fixed thresholds.

2.1 Existing Work

Existing health-monitoring systems mainly include conventional medical instruments, standalone wearable devices, and IoT-based monitoring systems. Conventional devices such as ECG machines and pulse oximeters provide accurate measurements but are mainly designed for clinical environments. Standalone wearable devices offer portability and easy access to health information, but many low-cost devices provide limited cloud storage, personalization, and intelligent analysis. IoT-based systems enable remote monitoring by transmitting health data to cloud platforms. However, many existing systems focus mainly on data collection and visualization rather than combining multiple physiological parameters with individual user information. Threshold-based systems may also generate unnecessary warnings due to differences in health conditions among individuals. Therefore, there is a need for an integrated system that combines wearable multi-parameter sensing, OLED visualization, cloud monitoring, user-profile information, and machine-learning-based health-status prediction.

2.2 Problems with the Existing Systems

- Limited continuous monitoring Conventional medical devices are mainly used in hospitals or clinical environments.
- Lack of portability: Traditional monitoring equipment is not convenient for everyday or continuous use.
- Limited parameters: Many wearable systems monitor only one or a few health parameters.
- Manual interpretation: Users often need to interpret sensor readings themselves.
- Dependence on fixed thresholds: Threshold-based systems may not consider individual differences between users.
- Limited personalization: Existing systems often do not combine health readings with user-specific information.
- Limited intelligent analysis: Many systems mainly display or store data without advanced machine-learning-based prediction.
- insufficient integration: Existing solutions may lack a single platform combining multi-sensor monitoring, local display, cloud storage, user profiles, and ML-based health-status prediction.

2.3 Gap Identification

- High power consumption for wearable devices
- Limited on-chip noise filtering.
- Lack of real-time ECG processing hardware
- Poor scalability and flexibility
- Missing hardware security features.
- CMOS scaling reliability challenges.

Chapter 3

Methodology

- **System Design and Overall Architecture :**

  The proposed system is an ESP32-based Smart Health Monitoring Belt that integrates ECG, SpO_2, and temperature sensors for continuous health monitoring. The ESP32 acts as the central controller to collect, process, and validate the sensor data. The measured parameters are displayed in real time on an OLED display for easy access. Using built-in Wi-Fi, the ESP32 transmits the collected data to the ThingSpeak cloud for storage and graphical visualization. A web application is developed to maintain user profiles and display health information. The sensor data and user-profile parameters are combined and provided to a machine-learning model for predicting the overall health status.

- **Wearable Belt and Sensor Placement :**

  The system uses a comfortable wearable belt or elastic strap to securely hold the electronic components and maintain proper sensor contact with the user's body. The ECG electrodes connected to the AD8232 module are positioned appropriately to obtain a measurable cardiac signal, while the MAX30102 sensor is placed with stable contact for SpO_2 and heart-rate measurement. The temperature sensor is positioned to obtain a representative body-temperature reading. The ESP32, battery, power-management circuit, and OLED display are mounted in a protected section of the belt. The design focuses on user comfort, mechanical stability, and protection of the electronic components. Proper sensor placement and stability are tested before continuous monitoring to minimize motion-related artifacts and

measurement errors.

- **ECG Data Acquisition**

  The AD8232 ECG sensor is used to acquire the electrical activity of the heart through ECG electrodes. The sensor conditions and amplifies the ECG signal before sending it to the ESP32 for processing. The ESP32 samples the signal and checks it for noise, abnormal values, and signal loss. Digital filtering techniques can be applied to reduce noise and interference. Important features such as heart rate, R-peaks, and RR intervals are extracted from the processed ECG signal. These features can be used as inputs for the machine-learning model.

- SpO_2 **and Heart-Rate Measurement** The MAX30102 sensor is used to measureSpO_2 and pulse-related information using red and infrared light. It communicates with the ESP32 through the I^2C interface and continuously provides optical sensor data. The ESP32 processes the raw signals to calculate SpO_2 percentage and heart rate. Signal averaging and filtering can be used to reduce fluctuations caused by movement or unstable contact. The measured values are displayed on the OLED and included in the cloud data. If the signal quality is insufficient, the system avoids generating an unreliable health prediction. Pasted markdown

- **Body-Temperature Measurement**

  A DS18B20 digital temperature sensor is incorporated into the belt to measure body temperature. The ESP32 reads and checks the temperature values for missing or unrealistic readings. A smoothing technique can be applied to reduce sudden fluctuations in the measurements. The temperature is displayed in degrees Celsius on the OLED and transmitted to ThingSpeak. The same data is also provided to the machine-learning pipeline for health-status analysis. Consistent sensor placement and measurement conditions are maintained during testing.

- **ESP32 Data Processing and Sensor Integration** The ESP32 coordinates all sensors and communication with external services. It initializes the ECG, MAX30102, temperature sensor, OLED display, and Wi-Fi connection during startup. The sensors are continuously read according to their required sampling rates, and

the ESP32 performs basic preprocessing and validation. The processed data includes ECG features, heart rate, SpO_2, temperature, and timestamp. This data is organized into a common format for OLED display, cloud transmission, and ML analysis.

- **OLED Display Implementation** A 0.96-inch SSD1306 OLED display is connected to the ESP32 through the I^2C interface for real-time monitoring. It displays parameters such as heart rate,SpO_2, temperature, ECG information, and health status. The display can also show Wi-Fi connection status, sensor errors, and cloud-upload information. Multiple screens can be used to present different parameters clearly. If a risk condition is detected, the OLED can display a warning message to the user. Pasted markdown

- **Wi-Fi Communication and ThingSpeak Cloud Integration** The ESP32 uses its built-in Wi-Fi to connect the smart belt to the Internet and communicate with the ThingSpeak cloud platform. Health parameters such as heart rate, SpO_2, temperature, and ECG-derived values are periodically uploaded to ThingSpeak. The cloud stores the data with timestamps and provides graphical visualization for monitoring health trends. Historical data can be analyzed to identify gradual changes or unusual patterns. If the Internet connection fails, the system continues local monitoring and periodically attempts to reconnect. Pasted markdown

- **Web Application and User Profile Development**

  A web application is developed for user registration, profile management, health-data visualization, and health-status monitoring. User details such as age, gender, height, and weight are stored and combined with sensor data for personalized ML analysis. The dashboard displays current measurements, historical graphs, and predicted health status. The application connects the wearable device, cloud data, user profile, and machine-learning system. Authentication and access control are included to protect user health information.

- **Dataset Preparation for machine learning**

  A suitable dataset containing ECG features, heart rate, SpO_2, temperature, and user-profile parameters is prepared for machine-learning analysis. The dataset is

checked for missing values, duplicate records, invalid measurements, and inconsistent labels. The health-status classes can include Normal, Attention Required, and Risk. Numerical data is normalized or standardized when required, while categorical data is converted into a suitable machine-readable format. Finally, the dataset is divided into training and testing sets for ML model development and evaluation.

- **Integration of User Profile with Sensor Data** The real-time physiological measurements are combined with user-profile information stored in the web application. Parameters such as age, gender, height, weight, heart rate, SpO_2, temperature, and ECG features are used to form the prediction input. The same preprocessing method used during model training is applied to the incoming data. The combined data is then provided to the trained ML model to generate a personalized health-status prediction.

- **Real-Time Health-Status Prediction** The trained machine-learning model receives the latest valid sensor readings along with the user's profile information. After pre processing, the model predicts a predefined health category such as Normal, Attention Required, or Risk. The predicted status is displayed on the web dashboard and can also be uploaded to ThingSpeak. Multiple measurements can be used to reduce false alerts caused by noisy readings

- **Cloud-Based Historical Analysis** ThingSpeak stores the collected health measurements with timestamps and provides graphical visualization of ECG-derived values, heart rate, SpO_2, and temperature. The historical graphs help identify gradual changes, repeated fluctuations, and unusual health patterns. The web application retrieves and presents this information through a user-friendly dashboard. This combination of real-time and historical data supports continuous health observation.

- **Alert and Abnormal-Condition Handling** The system generates an alert when sensor readings or ML predictions indicate a potentially abnormal condition. It combines sensor-validity checks, predefined thresholds, and ML results to improve reliability. If abnormal patterns persist, the OLED can display a warning and an LED or buzzer can provide a local alert. The web application can record the event

and, if required, notify an authorized caregiver. The alert logic is designed to reduce false alarms caused by sensor errors, movement, or poor contact.

- **System Testing and Validation**

  The complete system is tested in stages to verify hardware, software, communication, and ML performance. Each sensor is tested independently for stable readings, while the OLED, Wi-Fi, and ThingSpeak communication are also verified. All sensors are then operated together to identify possible conflicts and the web application is tested for profile management and prediction functions. Finally, the ML model is evaluated using an independent dataset and suitable and performance metrics. The complete belt is tested under controlled movement and normal operating conditions

- **Complete Operational Workflow** During operation, the user wears the belt and the ESP32 initializes the sensors and Wi-Fi connection. ECG, SpO_2, heart rate, and temperature data are continuously collected, processed, and displayed on the OLED. The processed data is uploaded to ThingSpeak and combined with the user's profile information through the web application. The trained ML model analyzes the combined data and generates a health-status classification. If an abnormal pattern is detected, an alert can be generated, and the complete monitoring process continues continuously.

3.1 Requirement Analysis

The proposed system requires a wearable belt or elastic strap to securely hold the sensors and electronic components while maintaining proper contact with the user. The main hardware requirements include an ESP32 development board, AD8232 ECG sensor, MAX30102 SpO_2 and heart-rate sensor, DS18B20 temperature sensor, and 0.96-inch SSD1306 OLED display. A rechargeable battery, charging module, voltage regulator, ECG electrodes, connecting wires, and protective casing are also required. The ESP32 must provide sensor interfacing, data processing, and built-in Wi-Fi connectivity for cloud communication. The system requires ThingSpeak for storing and visualizing health data remotely. A web application is required for user registration, profile management, health-data visualization, and health-status display. The machine-learning system

requires physiological and user-profile data for training and prediction. Software requirements include Arduino IDE, Embedded C/C++, Python, Pandas, NumPy, Scikit-learn, and web-development technologies such as HTML, CSS, JavaScript, and Bootstrap. The system should support real-time monitoring, data validation, cloud storage, ML-based health-status prediction, and abnormal-condition alerts. Finally, the complete system requires testing of sensors, OLED display, Wi-Fi communication, cloud integration, web application, and ML performance. [**?**].

3.1.1 Functional Requirements

- **Hardware Requirements**

- ESP32 Development Board – Main controller for sensor acquisition, processing, Wi-Fi communication, and ML-related data handling

- AD8232 ECG Sensor Module – For acquiring the electrical activity of the heart.

- MAX30102 Sensor – For measuring SpO_2 and heart rate/pulse rate.

- Body Temperature Sensor – DS18B20 or another suitable contact/non-contact temperature sensor for measuring body temperature.

- OLED Display (0.96-inch I^2C, SSD1306) – For displaying real-time sensor readings and health-status information.

- Li-ion/Li-Po Rechargeable Battery – Portable power supply for the wearable belt.

- Battery Charging Module – For safe charging of the rechargeable battery.

- Voltage Regulator/Power Management Module – To provide appropriate voltage levels to the ESP32 and sensors.

- Smart Belt/Elastic Wearable Strap – To securely mount the sensors and electronics on the user's body.

- Jumper Wires/PCB/Connectors – For electrical connections between the ESP32 and sensors.

- Buzzer/LED (Optional) – For providing an alert when abnormal health conditions are detected.
- Enclosure/Protective Casing – To protect the electronics and improve wearable safety.
- **software Requirements**
- Arduino IDE – For programming and uploading firmware to the ESP32.
- Embedded C/C++ – Programming language used for ESP32 firmware development.
- ESP32 Board Package – Required to compile and upload programs to the ESP32.
- Adafruit SSD1306 / Adafruit GFX Libraries – For controlling the OLED display
- MAX3010x Sensor Library – For interfacing the MAX30102 sensor and obtaining SpO2 / heart-rate data
- AD8232 ECG Interface Code/Signal Processing – For acquiring and preprocessing ECG signals.
- Wi-Fi Libraries for ESP32 – For connecting the device to the Internet.
- ThingSpeak Cloud Platform – For uploading, storing, visualizing, and monitoring sensor data remotely

3.1.2 Non-functional Requirements

- Small Chip Area

The design should occupy minimum silicon area. Compact circuits improve portability.

- High Speed Processing

Real-time ECG analysis requires fast processing.

- Scalability

Future upgrades such as AI-based diagnosis should be possible.

- Security

Patient health data should be protected during storage and transmission.

Table 3.1: Sample Requirement Specification

Requirement Type	Description
Functional	User authentication with email/OTP verification
Functional	Task scheduling with notifications
Non-functional	Should work on devices with minimum 2GB RAM
Non-functional	Cloud backup and data synchronization

3.2 System Design

The proposed system is designed as an ESP32-based Smart Health Monitoring Belt that integrates wearable sensing, local display, wireless communication, cloud storage, web application, and machine learning. The ECG, MAX30102, and temperature sensors continuously acquire physiological parameters from the user. The ESP32 acts as the central controller and performs data acquisition, basic processing, validation, and feature extraction. The processed values such as ECG, heart rate, SpO_2, and temperature are displayed on the OLED for real-time monitoring. The ESP32 uses its built-in Wi-Fi to transmit the collected data to the ThingSpeak cloud platform for storage and graphical visualization. A web application maintains the user's profile and provides a dashboard for viewing current and historical health information. The profile parameters are combined with real-time sensor measurements to create the input for the machine-learning model. The ML model analyzes the data and predicts a health-status category such as Normal, Attention Required, or Risk. If an abnormal pattern is detected, the system can provide a warning through the OLED, LED, or buzzer. The complete design follows the flow of sensor acquisition → ESP32 processing → OLED display → ThingSpeak cloud → web application → ML analysis → health-status prediction.

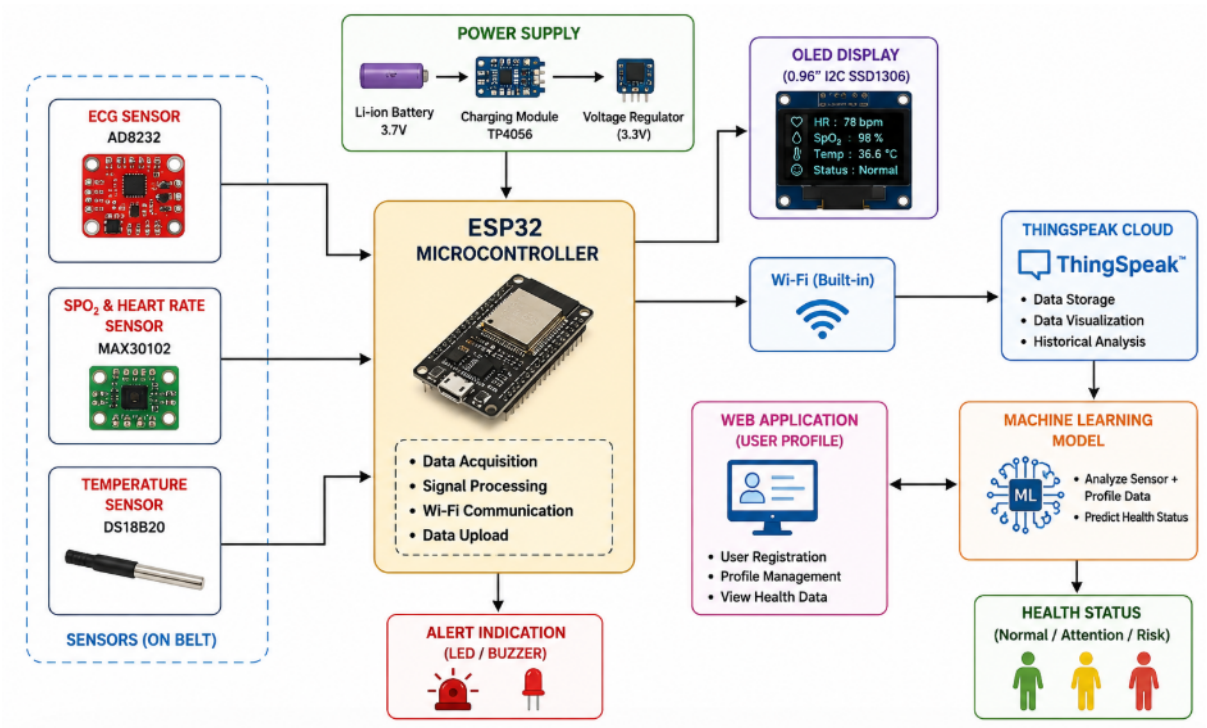


Figure 3.1: Block diagram of IOT and Machine learning Based Smart Health Monitoring Belt with ECG using ESP32

3.2.1 System Architecture

- ECG Signal Acquisition Electrodes are attached to the patient's body to capture heart electrical signals. The weak ECG signals are collected from the chest leads.
- Analog Front-End (AFE) Amplifies the low-voltage ECG signals using instrumentation amplifiers. Removes noise and interference through filters.
- Signal Conditioning Unit Includes low-pass, high-pass, and notch filters. Eliminates baseline wander, muscle noise, and power-line interference.
- Analog-to-Digital Converter (ADC) Converts analog ECG signals into digital form for processing.Enables accurate digital analysis inside the system.
- Memory Unit Stores patient ECG data temporarily or permanently. Helps in later medical analysis and monitoring.
- Wireless Communication Module Transfers ECG data through Bluetooth, Wi-Fi, or IoT modules. Enables remote patient monitoring.
- Display and User Interface Shows ECG waveform, heart rate, and alert messages. Can be connected to mobile apps or monitoring devices.
- Power Management System Uses battery-efficient circuits for portable operation.Ensures low power consumption in wearable devices.
- Alert and Diagnostic System Generates alerts during abnormal heart conditions. Assists doctors in early diagnosis and emergency monitoring.
- Cloud or Data Storage Integration Stores ECG records in cloud databases or hospital servers.Allows doctors to access patient data remotely.
- Output Monitoring Section Final processed ECG signal is displayed or transmitted.Provides continuous real-time cardiac monitoring.Final processed ECG signal is displayed or transmitted.Provides continuous real-time cardiac monitoring.

3.3 Algorithms Used

- ECG signal processing:Filtering, R-peak detection and RR-interval extraction for ECG analysis.
- SpO_2 and Heart Rate Algorithm: Processes red/IR signals from MAX30102 to calculate SpO_2 and heart rate.
- Moving Average Filtering: Smooths temperature and sensor readings by reducing sudden fluctuations.
- Data Validation: Checks missing, invalid, noisy, or out-of-range sensor values.
- Feature Engineering: Extracts useful features such as ECG parameters, heart rate, SpO_2 , temperature, age, height and weight.
- Random Forest Classification: Main ML algorithm used to classify health status as Normal, Attention Required, or Risk.
- Threshold-Based Alert: Detects abnormal readings and generates warnings through OLED..

3.4 Tools and Technologies

- ESP32 Tools Used for implementation and testing of ECG processing modules.Supports real-time signal processing.
- ML Collects ECG sensor data and interfaces with other modules.Controls data transmission and monitoring.
- ECG Sensors and Electrodes Capture electrical activity of the heart from the human body.Provide input signals to the system.Bluetooth Module (HC-05 / HC-06) Enables wireless ECG data transmission.Used for remote health monitoring.
- IoT Technology Connects ECG devices to cloud platforms for remote access Supports telemedicine applications.

- Proteus Simulation Software Simulates microcontroller and electronic circuit designs.Tests hardware functionality before implementation.

- DSP (Digital Signal Processing) Techniques Used for filtering, feature extraction, and noise reduction.

- CMOS Technology Used in ESP32 chip fabrication for low-power operation. Suitable for portable ECG devices.

Chapter 4

Implementation and Testing

4.1 System Implementation

The system is implemented using Flutter for cross-platform development, with Firebase as the backend database [**?**]. Dart is used as the programming language, while Firebase Authentication and Firestore provide secure login and data storage. The gamification module includes badges, rewards, and streak counters to enhance user engagement [**?**].

4.2 System Testing

Testing ensures that the developed application meets the specified requirements. Unit testing was conducted for individual modules, followed by integration testing to validate data flow between components. Finally, usability testing was performed with a group of students to evaluate user satisfaction and ease of use.

4.2.1 Unit Testing

4.2.2 Integration Testing

Chapter 5

Results and Discussion

The application successfully improves task completion rates and engagement compared to existing apps. Screenshots demonstrate the task scheduling interface and gamified feedback system.

5.1 Output Screenshots with Explanations

5.2 Comparison with Existing Systems

The proposed system, **Smart Task Management System with Gamification for Students**, was evaluated based on both system performance and user feedback. A pilot study was conducted with 30 students from the Department of Electronics and Communication Engineering, PESITM, Shivamogga.

The evaluation focused on four key aspects: (i) ease of use, (ii) accuracy of task scheduling, (iii) effectiveness of gamification features, and (iv) overall user satisfaction. Students were asked to use the system for two weeks and then complete a structured feedback form. Scores were collected on a scale of 1 to 5, where 1 indicates poor performance and 5 indicates excellent performance.

The results indicate that the system was highly effective in enhancing productivity and motivating students through gamification. The average rating across all parameters was **4.48/5**, which demonstrates the usefulness of the application in real-world academic scenarios.

In particular, the highest rating of 4.7 was obtained for *Overall User Satisfaction*,

Table 5.1: Sample Results of User Evaluation Metrics

Sl. No.	Evaluation Parameter	Average Score (out of 5)
1	Ease of Use (User Interface)	4.6
2	Task Scheduling Accuracy	4.4
3	Effectiveness of Gamification Features	4.2
4	Improvement in Time Management	4.5
5	Overall User Satisfaction	4.7

showing that students found the system engaging and beneficial. The comparatively lower score of 4.2 for *Gamification Features* suggests that while gamification motivated users, there is scope for enhancing this module by adding more diverse rewards and challenges.

5.3 Performance Evaluation

Chapter 6

Conclusion and Future Enhancements

The project demonstrates that gamification can significantly enhance task management for students. Future enhancements may include AI-based personalized reminders, advanced analytics, and integration with wearable devices.

6.1 Conclusion

The project **"IOT and Machine learning Based Smart Health Monitoring Belt with ECG using ESP32"** successfully demonstrated the ESP32-Based Smart Health Monitoring Belt is a portable system designed for continuous monitoring of physiological parameters. It integrates an ESP32, ECG sensor, SpO_2 sensor, temperature sensor, and OLED display to collect, process, and display real-time health information. The collected data is transmitted through Wi-Fi to the ThingSpeak cloud for storage and visualization. A web application maintains user profile information such as age, gender, height, and weight. These profile details are combined with sensor data and provided to a machine-learning model for personalized health-status prediction. The system classifies the user's condition into Normal, Attention Required, or Risk/Critical categories. Overall, the project integrates embedded systems, wearable technology, IoT, cloud computing, web applications, and machine learning for continuous health monitoring.

This outcome directly contributes to the United Nations Sustainable Development Goal (SDG) 4: Quality Education, as it fosters effective learning habits, enhances student engagement, and creates a supportive digital ecosystem that encourages continuous improvement in education.

6.2 Limitations

Although the system achieves its intended objectives, there are certain limitations:

- Sensor Accuracy: ECG and SpO_2 readings can be affected by movement, poor sensor contact, and motion artifacts.
- Limited Battery Life: Continuous sensing, OLED operation, and Wi-Fi communication can increase power consumption.
- Internet Dependency: Cloud monitoring and remote data access depend on a stable Wi-Fi/Internet connection.
- ML Dataset Dependency: The accuracy of health-status prediction depends on the quality, size, and diversity of the training dataset.
- Not Medical Grade: The system is an academic prototype and cannot replace clinically validated medical equipment or professional diagnosis.

6.3 Future Enhancements

To further improve the system, the following enhancements can be considered:

- Additional Sensors: Add blood-pressure, respiratory-rate, motion, and fall-detection sensors for more comprehensive monitoring.
- Advanced ML Models: Implement LSTM, CNN, or hybrid CNN-LSTM models for improved ECG and time-series analysis.
- Mobile Application: Develop a mobile app for real-time health monitoring, notifications, and historical data access..

- Emergency Alerts: Add GPS, SMS, email, or caregiver notifications for serious abnormal conditions.
- Improved Wearability: Use flexible PCBs, conductive fabric electrodes, and optimized sensor placement for better comfort and reliability.
- Clinical Validation: Test the system with larger, diverse, and clinically validated datasets to improve accuracy and reliability. ECG Health

6.4 Bibilography

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Appendix A

Figures/Tables/ Screen Shots

[Insert your appendix content here]

Appendix B

Publications

[Publication Details/Certicate/Patents]